

↳ We want to combine multiple tables into one single table.

↳ We join tables based on a common column between the different tables.

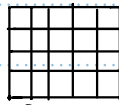
Data Warehouse



- We have different tables that are related to one another based on a specific column (primary/foreign key)



Write SQL code



- Summarised
- Joined
- Aggregated

• How we join tables

INNER JOIN

↳ Joins tables & returns rows that only have matching values on both tables

Syntax:

```
SELECT col 1,
        col 2,
        col 3,
FROM table 1 AS A
INNER JOIN table 2 AS B
ON A.common_column = B.common_column;
```

Eg.

customers

customer ID	customer Name
1	Ihabo
2	Sarah
3	Lerato
4	John

Orders

order ID	customer ID	Product
101	1	Laptop
102	2	Phone
103	2	Headphones
104	5	tablet

```
SELECT customer ID
        customer Name
        order ID
        product
FROM customers AS A
INNER JOIN orders AS B
ON A.customer ID = B.customer ID;
```

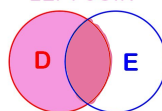
customer ID	customer Name	order ID	Product
1	Ihabo	101	Laptop
2	Sarah	102	Phone
2	Sarah	103	Headphones

Left join

Syntax:

```
SELECT col 1,
        col 2,
        col 3,
FROM table 1 AS A
LEFT JOIN table 2 AS B
ON A.common_column = B.common_column;
```

LEFT JOIN



- Returns all the rows from the left table and only

the matching rows from the right table

customers

customer ID	customer Name
1	Ihabo
2	Sarah
3	Lerato
4	John

Orders

order ID	customer ID	Product
101	1	Laptop
102	2	Phone
103	2	Headphones
104	5	tablet

SELECT customer ID,
customer Name,
product
FROM customers AS A
LEFT JOIN orders AS B
ON A.customer ID = B.customer ID

customer ID	customer Name	Product
1	Ithabo	Laptop
2	Barah	Phone
2	Barah	Head-phones
3	Lerato	Null
4	John	Null

Dimension table

↳ Descriptive table

↳ Static

↓
user - profile

↳ 5k rows

① customer

Fact table

↳ Analysis

↳ Continuously changes

↓
Viewership

↳ 50k rows

↳  multiple records

